

Apply Filters to SQL Queries

Project Description

In this project, I used SQL filtering techniques to investigate potential security concerns involving employee login activity and organisational devices. By querying the *'log_in_attempts'* and *'employees'* tables, I retrieved targeted information to support security investigations and identify systems needing updates. This activity shows how SQL can be used in cybersecurity to analyse data, investigate suspicious behaviour, and support security operations.

Retrieve After Hours Failed Login Attempts

A potential security incident was identified involving login activity occurring after normal business hours. To investigate this activity, I filtered the *'log_in_attempts'* table to locate failed login attempts occurring after 18:00.

Query used:

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_time > '18:00'
-> AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	aestrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

19 rows in set (0.121 sec)

This query uses the 'AND' operator to apply multiple filtering conditions at the same time. The first condition filters for login attempts occurring after 18:00, while the second condition only shows the results to failed login attempts using the 'success' field.

Filtering failed after hours activity is important because unsuccessful login attempts outside normal operating hours may indicate unauthorised access attempts, password guessing, or malicious behaviour needing further investigation.

Retrieve Login Attempts on Specific Dates

The security team identified suspicious activity occurring on 2022-05-09 and needed additional context around the event. To support the investigation, I retrieved login attempts occurring on both 2022-05-09 and the previous day.

Query used:

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_date = '2022-05-08'  
-> OR login_date = '2022-05-09';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arvise	2022-05-09	06:48:38	MEXICO	192.168.171.192	1

This query uses the 'OR' operator to return records matching either date. Reviewing activity across multiple dates helps investigators identify patterns, compare user behaviour, and decide if suspicious activity occurred over a longer period.

Date based filtering is commonly used during security investigations to narrow large datasets and focus analysis on a specific incident timeframe.

Retrieve Login Attempts Outside of Mexico

During the investigation, the security team confirmed that suspicious activity did not originate from Mexico. To narrow the investigation more, I filtered for login attempts originating outside of Mexico.

Query used:

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	graab	2022-05-11	01:45:14	CAN	192.168.170.243	1

This query combines ‘NOT’ and ‘LIKE’ filtering. The ‘LIKE’ keyword searches for patterns within the ‘country’ column, while the ‘%’ wildcard matches any characters following ‘MEX’. This allows the query to match both ‘MEX’ and ‘MEXICO’.

The ‘NOT’ operator does not include these results and returns login attempts originating from all other countries. Geographic filtering helps security teams focus investigations and identify potentially unusual access locations.

Retrieve Employees in Marketing

The security team needed to perform updates on employee devices belonging to the Marketing department within East building offices. To identify these systems, I filtered the ‘employees’ table by both department and office location.

Query used:

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Marketing'
-> AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267
1088	k865l965m233	rgosh	Marketing	East-157
1103	NULL	randerss	Marketing	East-460
1156	a184b775c707	dellery	Marketing	East-417
1163	h679i515j339	cwilliam	Marketing	East-216

7 rows in set (0.016 sec)

This query uses the 'AND' operator to make sure both conditions are met. Employees must belong to the Marketing department and have office locations beginning with 'East'.

The 'LIKE' keyword and '%' wildcard allow pattern matching for office locations such as 'East-170' and 'East-320'. This type of filtering helps administrators identify systems needing maintenance or security updates without reviewing the entire employee database.

Retrieve Employees in Finance or Sales

The organisation needed another security update targeting employees in the Finance and Sales departments. To locate these users, I filtered the employee records using multiple department conditions.

Query used:

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Finance'  
-> OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	ilansky	Finance	South-108

This query uses the 'OR' operator to return employees belonging to either department. Using 'OR' makes it possible to retrieve records across multiple categories in a single query.

Queries like this support operational security tasks by helping teams identify groups of devices or users needing planned updates or fixing.

Retrieve All Employees Not in IT

Employees in the Information Technology department had already received the needed update. To identify remaining systems needing attention, I did not include IT employees in the results.

Query used:

```

MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE NOT department = 'Information Technology';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1001 | b239c825d303 | bmoreno | Marketing | Central-276 |
| 1002 | c116d593e558 | tshah | Human Resources | North-434 |
| 1003 | d394e816f943 | sgilmore | Finance | South-153 |
| 1004 | e218f877g788 | eraab | Human Resources | South-127 |
| 1005 | f551g340h864 | gesparza | Human Resources | South-366 |
| 1007 | h174i497j413 | wjaffrey | Finance | North-406 |
| 1008 | i858j582k571 | shenard | Finance | South-170 |

```

This query uses the 'NOT' operator to not include employees belonging to the Information Technology department and return records from all remaining departments.

Exclusion filtering is useful during security operations because it helps narrow scope and avoid duplicate work during patching, system updates, or fixing issues.

Summary

In this project, I used SQL filtering techniques to investigate login activity and support device management decisions in an organisational environment. I applied 'AND', 'OR', 'NOT', and 'LIKE' operators to retrieve targeted records involving dates, times, geographic locations, departments, and office locations. This activity strengthened my understanding of SQL as a cybersecurity tool and showed how database queries support security investigations, access analysis, and operational decision-making.